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**BCA Semester-III  
CAUE302: Python Programming**

**Assignment-2**

**Sequence Types**

**"Instructions: Write and execute the following programs in Jupyter Notebook (24BCA001\_Assignment\_2.ipynb)."**

- 1. Accept a string from the user and convert it into Uppercase, Lowercase, and Title Case.**

| Sample Input                     | Expected Output                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------|
| Enter string: python programming | Uppercase: PYTHON<br>PROGRAMMING<br>Lowercase: python programming<br>Title Case: Python Programming |

- 2. Accept a string from the user and display its First character, Last character, and Total length using len().**

| Sample Input           | Expected Output                                      |
|------------------------|------------------------------------------------------|
| Enter string: CHARUSAT | First Character: C<br>Last Character: T<br>Length: 8 |

- 3. Create a list of 4 fruit names. Append a new fruit to the end using append() and print the updated list.**

| Sample Input                                                      | Expected Output                                      |
|-------------------------------------------------------------------|------------------------------------------------------|
| Initial List: ['apple', 'banana', 'mango']<br>Input fruit: cherry | Updated List: ['apple', 'banana', 'mango', 'cherry'] |

- 4. Accept 5 numbers from the user into a list. Calculate and print the Sum and Length of the list.**

| Sample Input                      | Expected Output             |
|-----------------------------------|-----------------------------|
| Input numbers: 10, 20, 30, 40, 50 | Total Sum: 150<br>Length: 5 |

5. Create a tuple containing 4 color names. Display the element at index 2 and print the index of 'blue' using index().

| Sample Input                              | Expected Output                                |
|-------------------------------------------|------------------------------------------------|
| Tuple: ('red', 'green', 'blue', 'yellow') | Element at Index 2: blue<br>Index of 'blue': 2 |

6. Accept a string from the user and display its Reverse using string slicing .

| Sample Input            | Expected Output            |
|-------------------------|----------------------------|
| Enter string: Education | Reversed String: noitacudE |

7. Accept a list of numbers from the user. Display the Largest and Smallest number using built-in max() and min() functions.

| Sample Input                      | Expected Output                          |
|-----------------------------------|------------------------------------------|
| Enter numbers: 45, 12, 89, 33, 67 | Maximum Number: 89<br>Minimum Number: 12 |

8. Accept a list of student marks. Sort the list in Ascending order using sort() and then in Descending order using sort(reverse=True).

| Sample Input                      | Expected Output                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------|
| Input Marks: [75, 42, 90, 68, 85] | Ascending Order: [42, 68, 75, 85, 90]<br>Descending Order: [90, 85, 75, 68, 42] |

9. Write a Python program to demonstrate Tuple Unpacking by storing Roll No, Name, and Course in a tuple and unpacking them into individual variables.

| Sample Input                        | Expected Output                                   |
|-------------------------------------|---------------------------------------------------|
| Tuple: (101, 'Krupa Pathak', 'BCA') | Roll No: 101<br>Name: Krupa Pathak<br>Course: BCA |

10. Accept a string from the user and count the total number of Vowels (a, e, i, o, u) present in it using a loop and count.

| Sample Input                        | Expected Output        |
|-------------------------------------|------------------------|
| Enter a string: Welcome to CHARUSAT | Total Vowels Count = 7 |

**11. Accept a string from the user and check whether it is a Palindrome (reads the same forward and backward).**

| Sample Input          | Expected Output              |
|-----------------------|------------------------------|
| Enter a string: madam | madam is a Palindrome string |

**12. Accept a list of integers from the user and separate them into two new lists: one for Even numbers and one for Odd numbers.**

| Sample Input                       | Expected Output                                                 |
|------------------------------------|-----------------------------------------------------------------|
| Input List: [12, 7, 15, 20, 9, 14] | Even Numbers List: [12, 20, 14]<br>Odd Numbers List: [7, 15, 9] |

**13. Write a Python program to remove all duplicate elements from a given list while maintaining the original order.**

| Sample Input                         | Expected Output                                  |
|--------------------------------------|--------------------------------------------------|
| Input List: [10, 20, 10, 30, 20, 40] | List after removing duplicates: [10, 20, 30, 40] |

**14. Write a Python program to input a password and check whether it meets the following criteria:**

- At least 8 characters long
- Contains at least one digit
- Contains at least one uppercase letter

**15. Write a Python program to find the average of all numbers stored in a tuple.**

**16. Write a Python program to find the second largest element in a tuple.**

**17. Create a dictionary containing five fruits and their prices. Print the complete dictionary.**

**18. Create a dictionary named subjects with 5 subject names as keys and marks as values. Print each subject with its marks using a loop.**

**19. Take input from the user for n students — ID and name — and store them in a dictionary. Then ask the user to search for a student by ID and display their name (or "not**

found" if the ID doesn't exist).

**20. Write a python program to create a dictionary with 5 Student id as keys and values as their 5 subject marks. find the percentages of each student and create a new dictionary to store percentage along with student id.  
e.g. {'23BCA001': [43,45,67,78,89]}  
output: {'23BCA001': 55}**

**21. Write a Python program to count the frequency of each character in a string using a dictionary.**

**Sample Input:**

Enter a string: python programming

**Sample Output:**

```
{  
'p': 2,  
'y': 1,  
't': 1,  
'h': 1,  
'o': 2,  
'n': 2,  
' ': 1,  
'r': 2,  
'g': 2,  
'a': 1,  
'm': 2,  
'i': 1  
}
```

**22. Store the monthly electricity bill of six months in a dictionary. Find the month with the highest bill.**

**Sample Input**

**Enter electricity bill for January: 2200**

**Enter electricity bill for February: 1850**

**Enter electricity bill for March: 2400**

**Enter electricity bill for April: 1990**

**Enter electricity bill for May: 2100**

**Enter electricity bill for June: 1950**

**Sample Output**

**Highest Electricity Bill**

**Month : March**

**Bill : ₹2400**

**23. Write a Python program to accept two sets of subjects chosen by two students and find:**

- **Common subjects**
- **Subjects only chosen by the first student**
- **Union of all chosen subjects**

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### **Sample Input**

```
Enter number of subjects chosen by Student 1: 4
Enter Subject 1: Python
Enter Subject 2: Java
Enter Subject 3: DBMS
Enter Subject 4: AI
```

```
Enter number of subjects chosen by Student 2: 4
Enter Subject 1: Python
Enter Subject 2: AI
Enter Subject 3: C
Enter Subject 4: Mathematics
```

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### **Sample Output**

```
Student 1 Subjects:
{'Python', 'Java', 'DBMS', 'AI'}
```

```
Student 2 Subjects:
{'Python', 'AI', 'C', 'Mathematics'}
```

```
Common Subjects:
{'Python', 'AI'}
```

```
Subjects only chosen by Student 1:
{'Java', 'DBMS'}
```

```
Union of all Subjects:
{'Python', 'Java', 'DBMS', 'AI', 'C', 'Mathematics'}
```

**24. Given a set of all roll numbers and another set of students present in class, write a Python program to find and display the roll numbers of absent students.**

Sample Input

Enter total number of students: 6

Enter all roll numbers:

101

102

103

104

105

106

Enter number of students present: 4

Enter present roll numbers:

101

103

104

106

Sample Output

All Roll Numbers:

{101, 102, 103, 104, 105, 106}

Present Students:

{101, 103, 104, 106}

Absent Students:

{102, 105}

**25. Write a Python function named `greet_user(name)` that accepts a person's name as an argument and prints a welcome message.**

| Sample Input / Call                    | Expected Output                               |
|----------------------------------------|-----------------------------------------------|
| Call: <code>greet_user('Krupa')</code> | Output: Welcome to Python Programming, Krupa! |

**26. Write a Python function `square_number(num)` that accepts a number and returns its square value.**

| Sample Input / Call                 | Expected Output     |
|-------------------------------------|---------------------|
| Call: <code>square_number(6)</code> | Output: Square = 36 |

**27. Write a Python program using the sorted() built-in function to sort a list of student names in both ascending and descending order.**

| Sample Input / Call                            | Expected Output                                                                                 |
|------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Input List: ['Krupa', 'Alice', 'David', 'Bob'] | Ascending: ['Alice', 'Bob', 'David', 'Krupa']<br>Descending: ['Krupa', 'David', 'Bob', 'Alice'] |

**28. Write a Python function calculate\_simple\_interest(principal, rate=7.5, time=1) using default arguments to calculate interest.**

| Sample Input / Call                                           | Expected Output                           |
|---------------------------------------------------------------|-------------------------------------------|
| Call 1:<br>calculate_simple_interest(10000)                   | Interest 1 = 750.0<br>Interest 2 = 2400.0 |
| Call 2:<br>calculate_simple_interest(10000, rate=8.0, time=3) |                                           |

**29. Write a Python function to Calculate the Body Mass Index (BMI) using height and weight.**

**Criteria:**

- **Input:** height (in meters), weight (in kg)
- **Formula:**

**BMI = weight / height<sup>2</sup>**

- **Output:** BMI value rounded to 2 decimal places (BMI = weight/height<sup>2</sup>)

**30. Write a Python function multiply\_all(\*args) using arbitrary positional arguments (\*args) to multiply any number of given integers.**

| Sample Input / Call           | Expected Output |
|-------------------------------|-----------------|
| Call 1: multiply_all(2, 3, 4) | Output 1 = 24   |
| Call 2: multiply_all(5, 10)   | Output 2 = 50   |

**31. Write a Python menu-driven program using user-defined functions for basic calculator operations (Add, Subtract, Multiply, Divide) that executes continuously until the user chooses to exit.**

| Sample Input / Call                     | Expected Output      |
|-----------------------------------------|----------------------|
| Select Choice: 1 (Add)   Inputs: 10, 20 | Result: 10 + 20 = 30 |

**32. Write a Python function `calculate_grade(marks_list)` that accepts a list of subject marks, calculates the percentage, and returns the Grade (A/B/C/F) along with pass/fail status.**

| Sample Input / Call                                      | Expected Output                               |
|----------------------------------------------------------|-----------------------------------------------|
| Call: <code>calculate_grade([80, 75, 90, 85, 70])</code> | Percentage: 80.0%   Grade: A   Status: PASSED |